Project 2 report – Menes Adisso

GitHub repo: https://github.com/menes-ux/ALU_Programing_in_C_Project_2.git

Project 1: Dynamic Academic Performance and Risk Analysis

1. Source code
2. Sample program output
3. The point of this analysis is to evaluate students' academic performance and identify those who may need additional support. The program examines each student's grade and places them into one of four categories:

Grade	Classification
80–100	Excellent
60–79	Satisfactory
50–59	Needs Improvement
Below 50	At Risk

The program counts the students in each category and calculates their percentage using:

$$\text{Category Percentage} = \frac{\text{Students in Category}}{\text{Total Students}} \times 100$$

Students scoring below 60 are identified as needing academic support.

4. Output removing and adding student

Project 2: Dynamic Text Mining and Linguistic Pattern Analyzer

1. Source code
2. Paragraph (*Index of NineteenEightyFour-Novel-GeorgeOrwell/*, n.d.)

   The hallway smelt of boiled cabbage and old rag mats. At one end of it a coloured poster, too large for indoor display, had been tacked to the wall. It depicted simply an enormous face, more than a metre wide: the face of a man of about forty-five, with a heavy black moustache and ruggedly handsome features. Winston made for the stairs. It was no use trying the lift. Even at the best of times it was seldom working, and at present the electric current was cut off during daylight hours. It was part of the economy drive in preparation for Hate Week. The flat was seven flights up, and Winston, who was thirty-nine and had a varicose ulcer above his right ankle, went slowly, resting several times on the way. On each landing, opposite the lift-shaft, the poster with the enormous face gazed from the wall. It was one of those pictures which are so contrived that the eyes follow you about when you move. BIG BROTHER IS WATCHING YOU, the caption beneath it ran.

   Book: Nineteen Eighty-Four (1984)
   Author: George Orwell
   Publication year: 1949
   Section: Part One, Chapter 1
   PDF page: 5
   Custom Analysis: Lexical Diversity Score

3. Sample output demonstrating the required operations and memory addresses.
4. The purpose of lexical diversity analysis is to measure vocabulary variety in a text.

   The program counts the total number of words and the number of distinct words, then calculates:

$$\text{Lexical Diversity (\%)} = \frac{\text{Distinct Words}}{\text{Total Words}} \times 100$$

The higher the percentage is, the more the words are different. Additionally, the score is also affected by text length, so it should not automatically determined if a text is better or not.

5. Evidence that the program handles punctuation, repeated words, and dynamically growing input correctly

Project 3: Dynamic IoT Event Monitoring and Callback Engine

1. Source code
2. Sample output demonstrating 12 readings simulated readings

3. The program uses 3 function pointers to display readings to the appropriate monitoring callbacks: temperature_monitor() for the temperature, pressure_monitor() for the pressure, and battery_monitor() for the battery voltage.

   Evidence of callback dispatch for all three required device types.

4. Evidence of device registration and removal.
5. The custom callback, `sensor_drift_detection()`, is used to identify sudden changes in sensor readings. It works by comparing the current reading with the previous one from the same device and calculating the difference.

   Change = |Current Reading − Previous Reading|

   If the difference is greater than 8°C for temperature, 12 kPa for pressure, or 0.40 V for battery voltage, the program generates an alert. For the first reading, there is nothing to compare it with, so it is used as the starting point. This helps us identify unusual changes in sensor readings that may need to be checked.

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6. Sample output showing normal and abnormal readings

Sources

Index of NineteenEightyFour-Novel-GeorgeOrwell/. (n.d.). Retrieved October 28, 2026, from https://dn790002.ca.archive.org/0/items/NineteenEightyFour-Novel-GeorgeOrwell/